

KKT condⁿ

$$\begin{array}{ll} \max / \min & f(x) \\ \text{s.t.} & x \in \mathbb{R}^k \end{array}$$

m equalities
means m eqⁿ's where $h(x) = 0$
 n inequalities
 $\hookrightarrow$ n equations where $g(x) \leq 0$

creating a lagrange function

$$L(x, \underbrace{\lambda_1, \dots, \lambda_m}_{\substack{\downarrow \\ m \text{ equalities}}}, \underbrace{\mu_1, \dots, \mu_n}_{\substack{\downarrow \\ n \text{ inequalities}}})$$

$$= f(x) + \sum_{i=1}^m \underbrace{\lambda_i h_i(x)}_{\substack{\downarrow \\ \text{for } m \text{ equalities}}} + \sum_{j=1}^n \underbrace{\mu_j g_j(x)}_{\substack{\downarrow \\ \text{for } n \text{ inequalities}}}$$

$$\min L(x, \lambda, \mu) \text{ w.r.t. } x \Rightarrow \nabla_x L(x, \lambda, \mu) = 0$$

differentiating the eqⁿ w.r.t x
Suppose u get K eqⁿ

then $\frac{\partial L}{\partial \lambda}$ we get m equations

$\frac{\partial L}{\partial \mu}$ we get n equations

eg

$$\text{Min } f(x, y) = (x-4)^2 + (y-4)^2$$

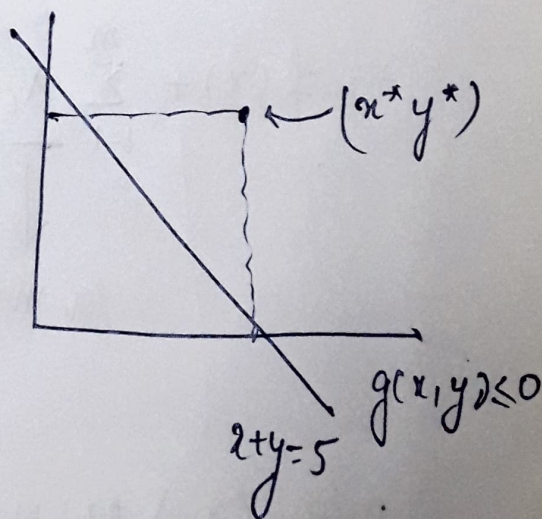
$$\text{s.t. } x+y \geq 5$$

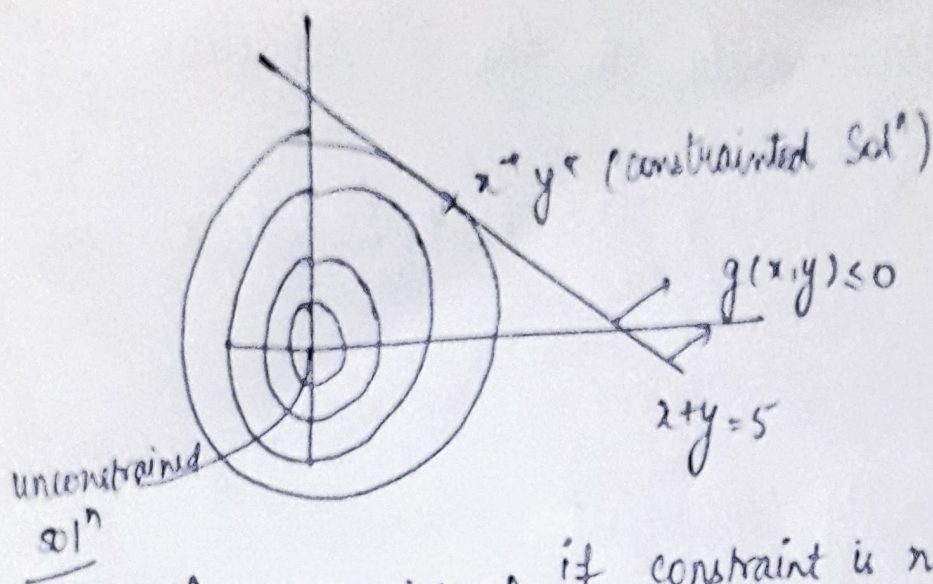
$$f(x, y) = (x-4)^2 + (y-4)^2$$

$$g(x, y) = -x - y + 5 \leq 0$$

$g(x, y) \leq 0$
constraint has no
part
 $\mu = 0$

$$\boxed{\mu g(x, y) = 0}$$





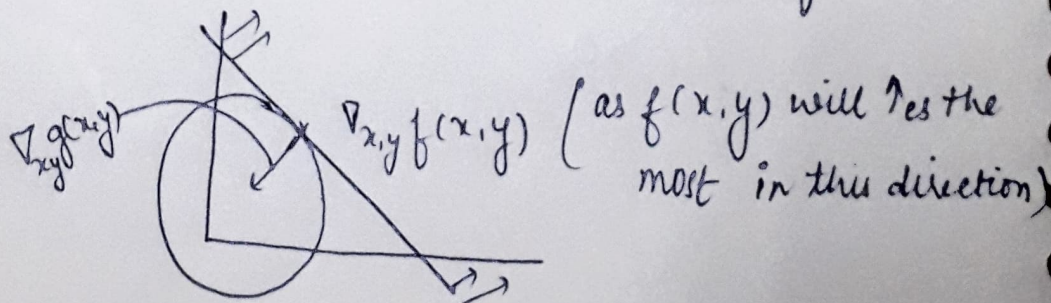
(or unconstrained & constraint soln are same) if constraint is not taking part in soln then generally μ part is $= 0$

Suppose if constraint is taking part in changing the solution then solution lies on $g(x, y) = 0$

then $\mu(g(x, y)) = 0$
where $\mu \neq 0$

Now,

$\mu_i \geq 0 \quad \forall i$ (why?)



$$\nabla_{x,y} L = 0$$

$$\Rightarrow \nabla f(x, y) + \mu \nabla g(x, y) = 0$$

$$\mu = -\nabla f(x, y) / \nabla g(x, y) \quad \boxed{\mu \geq 0}$$

(1) Stationarity is the 1st condition

$$\nabla_x f(x) + \sum_{i=1}^m \lambda_i h_i(x) + \sum_{i=1}^n \mu_i g_i(x) = 0$$

(Fotminimisation) .

(II) Equality Constraint

$$\nabla_x f(x) + \sum_{i=1}^m \lambda_i h_i(x) + \sum_{i=1}^n \mu_i g_i(x) = 0$$

$$\sum_{i=1}^m \nabla_x \lambda_i h_i(x) = 0$$

(III) Inequality constraint :-

(complementary slackness)

$$\mu_i g_i(x) = 0 \quad \forall i = 0, 1, \dots, n$$

$$\mu_i \geq 0$$

Kushn Tucker conditions

① calculation

① $\frac{\partial L}{\partial x_1} = 0$ ② $\frac{\partial L}{\partial x_2} = 0$

③ $\lambda (\text{constraint}) = 0$ ④ $h \leq 0$

⑤ $x_1, x_2 \geq 0$

$\lambda \geq 0$ for maxima

$\lambda \leq 0$ for minima

$L(x_1, x_2, \lambda)$

= objective funⁿ - λ (constraint)

(similar way we will

calculate the way we did it in lagrange

$\frac{\partial L}{\partial x_1} = 0$ ① $\frac{\partial L}{\partial x_2} = 0$ ②

$\lambda (\text{constraint}) = 0$ ④

constraint ≤ 0

$x_1, x_2 \geq 0$ $\lambda \geq 0$ ⑤

⑥ we need to get the value of z

case I $\lambda = 0$

case (II) $\lambda \neq 0$

we will only use $q^*(1)$, (11) & (11)

$$4x_1 + 12x_2 - 2\lambda = 0$$

$$4x_1 + 12x_2 = 0$$

$$-14x_2 + 12x_1 = 0$$

$$x_1 = a \quad x_2 = b$$

put in $Z = \infty$
invalid solⁿ

$$Z = 2x_1^2 - 7x_2^2 + 12x_1x_2$$

$$\text{sub to } 2x_1 + 5x_2 \leq 98$$

$$x_1, x_2 \geq 0$$